

Haoyang Hong

Second-year Ph.D. Student in Artificial Intelligence, Oregon State University

B.S. in Statistics, University of Science and Technology of China

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Research Interests

Reinforcement learning theory, bandit theory, online learning, KL-regularized learning, robustness, and statistical inference under adaptive data collection.

Education

Oregon State University

Ph.D. in Artificial Intelligence

Advisor: Huazheng Wang

Sep. 2024 – Present

University of Science and Technology of China

B.S. in Statistics, School of the Gifted Young

Sep. 2020 – Jul. 2024

Publications and Preprints

Online KL-Regularized Reinforcement Learning with Function Approximation under Misspecification

Haoyang Hong*, Zichen Wang*, Quanquan Gu, Huazheng Wang

RLC 2026

When Can You Poison Rewards? A Tight Characterization of Reward Poisoning in Linear MDPs

Jose Efraim Aguilar Escamilla*, Haoyang Hong*, Jiawei Li*, Haoyu Zhao, Xuezhou Zhang, Sanghyun Hong,

Huazheng Wang

ICML 2026

Beyond Static Bias: Adaptive Multi-Fidelity Bandits with Improving Proxies

Muyun Lu*, Haoyang Hong*, Huazheng Wang, Ying Lin

arXiv:2605.08558, 2026

Design-Based Bandits under Network Interference: A Trade-off between Regret and Inference

Zichen Wang*, Haoyang Hong*, Chuanhao Li, Haoxuan Li, Zhiheng Zhang, Huazheng Wang

NeurIPS 2025

*Equal contribution.

Selected Research Directions

Regularized and Robust Reinforcement Learning

- Develop theory for online KL-regularized reinforcement learning under function approximation and model misspecification, motivated by reference-policy regularization in modern post-training.
- Study instance-level vulnerability and robustness of reinforcement learning systems under structured reward perturbations.

Bandit Learning under Adaptive Data Collection

- Design online learning algorithms that balance regret minimization with valid statistical inference when data are collected adaptively.
- Study bandit models with network interference, preference feedback, forced exploration, and clean non-asymptotic guarantees.

Multi-Fidelity and Cost-Aware Online Learning

- Analyze adaptive multi-fidelity bandits with improving proxy feedback, where low-cost signals become more informative through repeated use.
- Build theory-guided simulation and evaluation pipelines for regret, robustness, fidelity allocation, and inference trade-offs.